Dynamic reporting for reproducible data analysis

A Workshop by the Center for Reproducible Science

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Objectives

- Recall different aspects of reproducibility
- Learn fundamental concepts of dynamic reporting
- Understand which problems dynamic reporting solves

Example

| Human Microbiome | Research Article

Major data analysis errors invalidate cancer microbiome findings

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ABSTRACT We re-analyzed the data from a recent large-scale study that reported strong correlations between DNA signatures of microbial organisms and 33 different cancer types and that created machine-learning predictors with near-perfect accuracy at distinguishing among cancers. We found at least two fundamental flaws in the reported data and in the methods: (i) errors in the genome database and the associated computational methods led to millions of false-positive findings of bacterial reads across all samples, largely because most of the sequences identified as bacteria were instead human; and (ii) errors in the transformation of the raw data created an artificial signature, even for microbes with no reads detected, tagging each tumor type with a distinct signal that the machine-learning programs then used to create an apparently accurate classifier. Each of these problems invalidates the results, leading to the conclusion that the microbiome-based classifiers for identifying cancer presented in the study are entirely wrong. These flaws have subsequently affected more than a dozen additional published studies that used the same data and whose results are likely invalid as well.

Gihawi et al. (2023)

Terminology

		Data	
		Same	Different
Analysis	Same	Reproducible	Replicable
	Different	Robust	Generalisable

The Turing Way Community (2022)

Generalizability

Reproducible	Replicable
Robust	Generalisable

Generalizeability is a multi-faceted concept, among other things referring to:

- **Extrapolation**: Extending conclusions available from studies in one or more subgroups of a population to another subgroup of the population
- **Translation**: Converting results in basic research into results that directly benefit humans

Generalizability: Example

Reproducible	Replicable
Robust	Generalisable

nature > technology features > article

TECHNOLOGY FEATURE | 09 January 2023 | Correction 12 January 2023

The reproducibility issues that haunt health-care AI

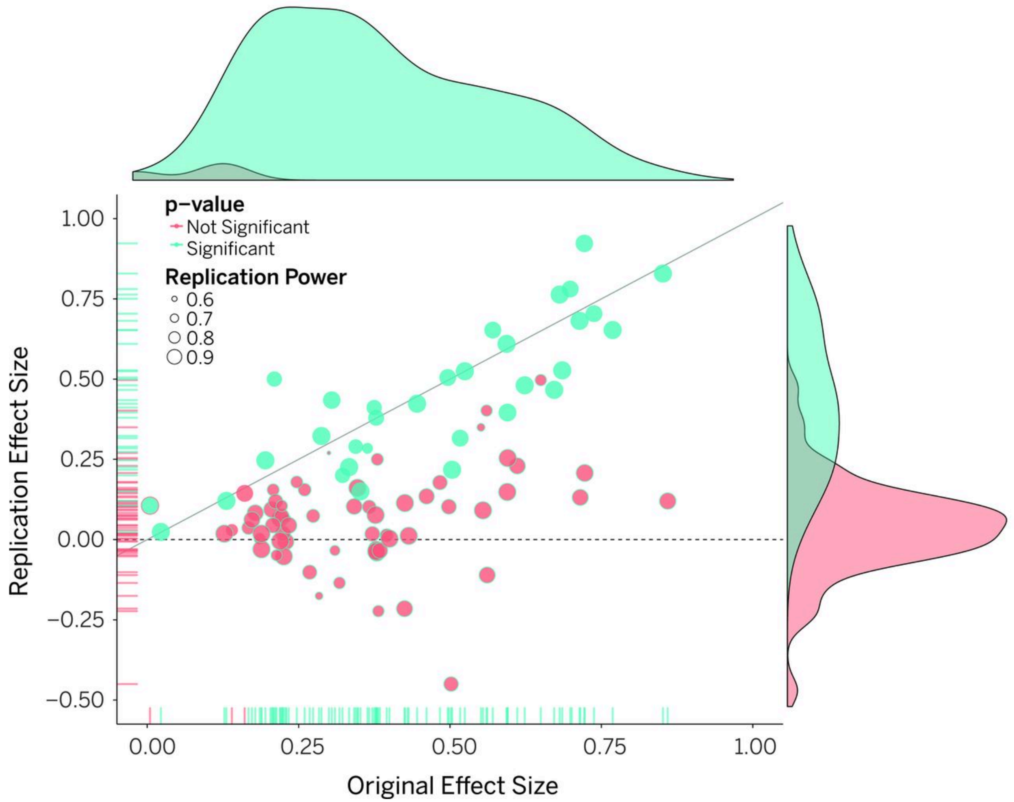
Health-care systems are rolling out artificial-intelligence tools for diagnosis and monitoring. But how reliable are the models?

> "Almost all of these award-winning models failed miserably. That was kind of surprising to us."

Sohn (2023)

Replicability: Example

Reproducible	Replicable
Robust	Generalisable



Open Science Collaboration (2015)

Robustness: Example

Reproducible	Replicable
Robust	Generalisable

REPRODUCIBILITY GOES ON TRIAL IN ECOLOGY

With same data sets, 246 biologists get different results – showing how analytical choices drive conclusions.

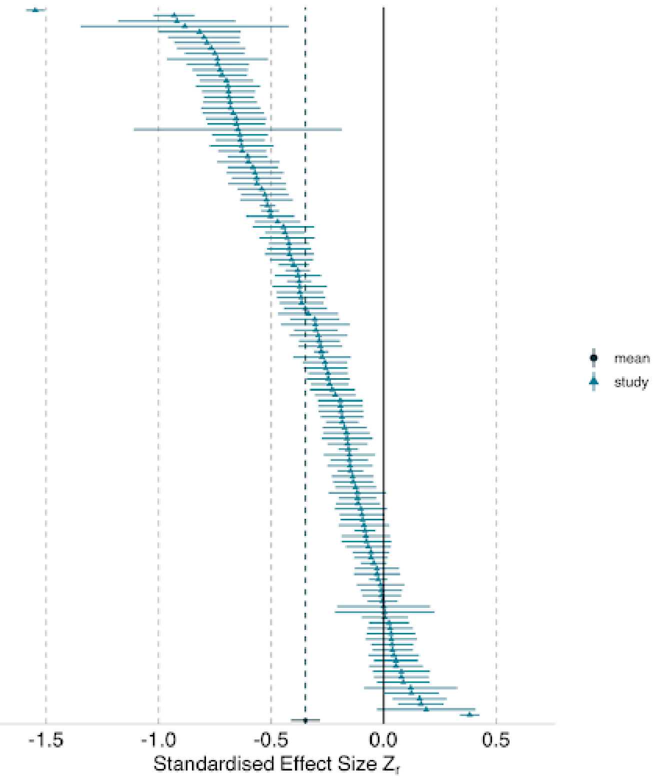
By Anil Oza

In a massive exercise to examine reproducibility, more than 200 biologists analysed the same sets of ecological data – and got widely divergent results. The first sweeping study of its kind in ecology demonstrates how much results in the field can vary, not because of differences in the environment, but because of scientists' analytical choices (E. Gould *et al*. Preprint at EcoEvoRxiv https://doi.org/gswhwr; 2023).

"There can be a tendency to treat individual papers' findings as definitive," says Hannah Fraser, an ecology meta researcher at the University of Melbourne in Australia and a co-author of the study. But the work shows that "we really can't be relying on any individual result or any individual study to tell us the whole story".

Variation in results might not be surprising, but quantifying that variation in a formal study could catalyse a larger movement to improve reproducibility, says Brian Nosek,

Nature | Vol 622 | 26 October 2023 | **677**



Gould et al. (2023)

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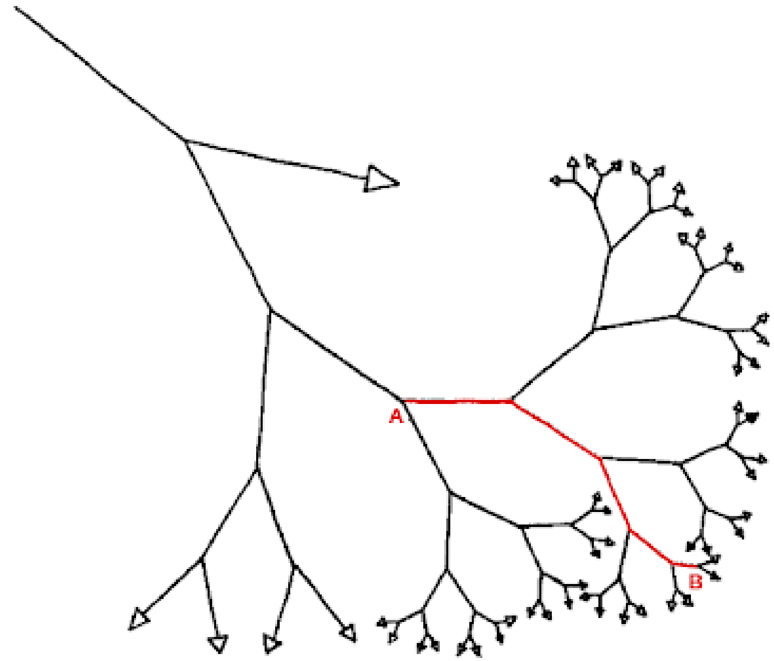
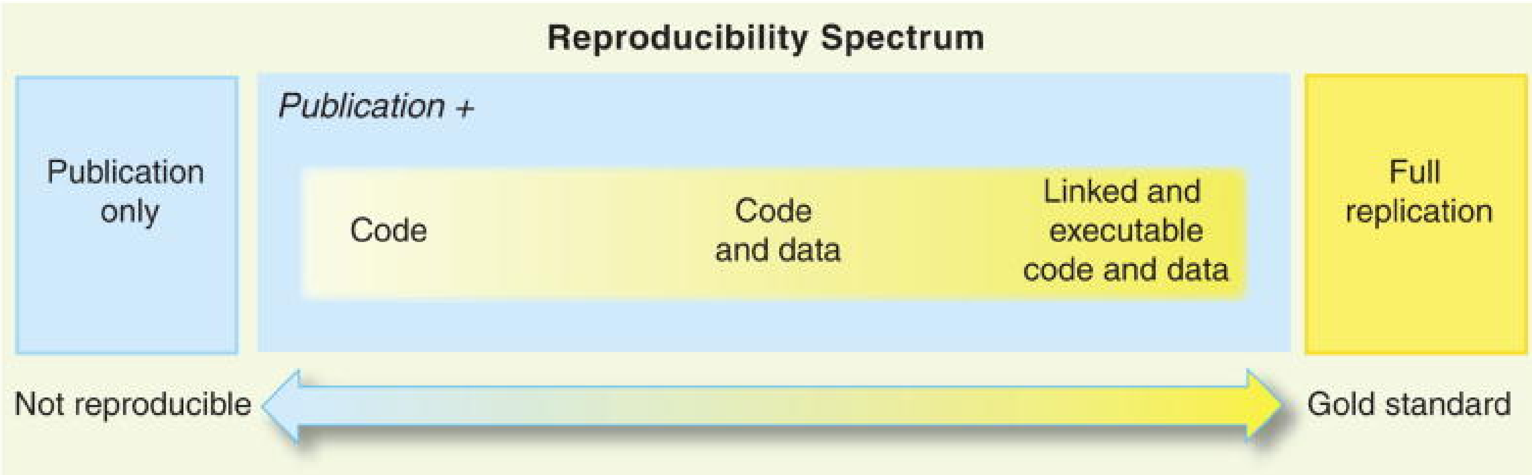
Threats to robustness and replicability

- "Garden of Forking Paths"
  Gelman and Loken (2013)
- "Researcher Degrees of Freedom"
  Simmons, Nelson, and Simonsohn (2011)
- "Selective reporting"
  Hoffmann et al. (2021)



Reproducibility

Reproducible	Replicable
Robust	Generalisable



Peng (2011)

Additional steps:

- versioning (e.g. tutorial at https://swcarpentry.github.io/git-novice/)
- containerization (e.g. CRS Primer Fraga González et al. (2024))
- managing dependencies with `make` (e.g. Peikert and Brandmaier (2019))

Reproducibility: typical problems

- cleandata.R
- data.csv
- plots.R
- stats.R
- tables.R

- unclear execution order
- probably poor documentation
- unclear further processing of output

Dynamic reporting

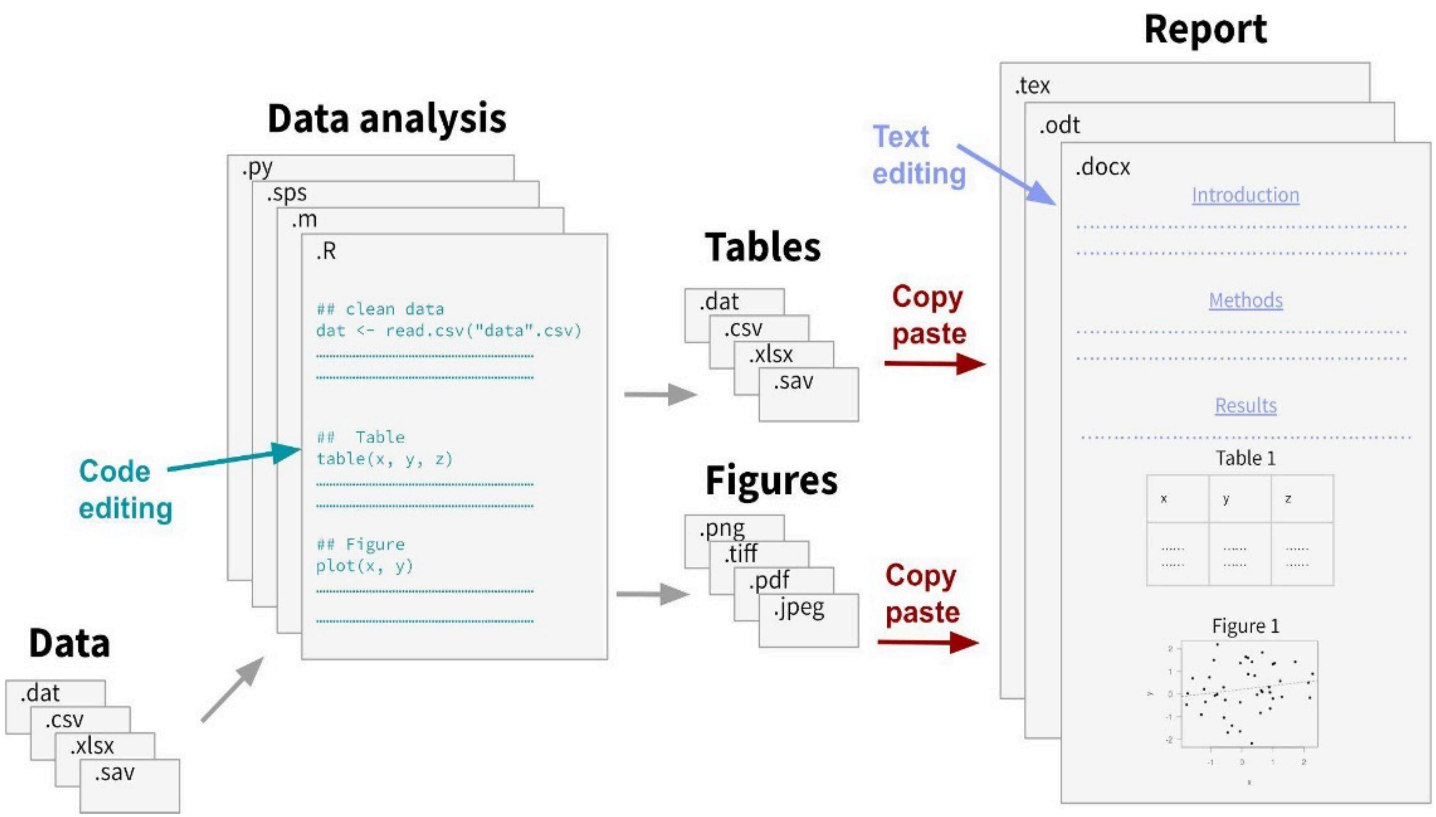
Problems

- unclear execution order
- probably poor documentation
- unclear further processing of the output

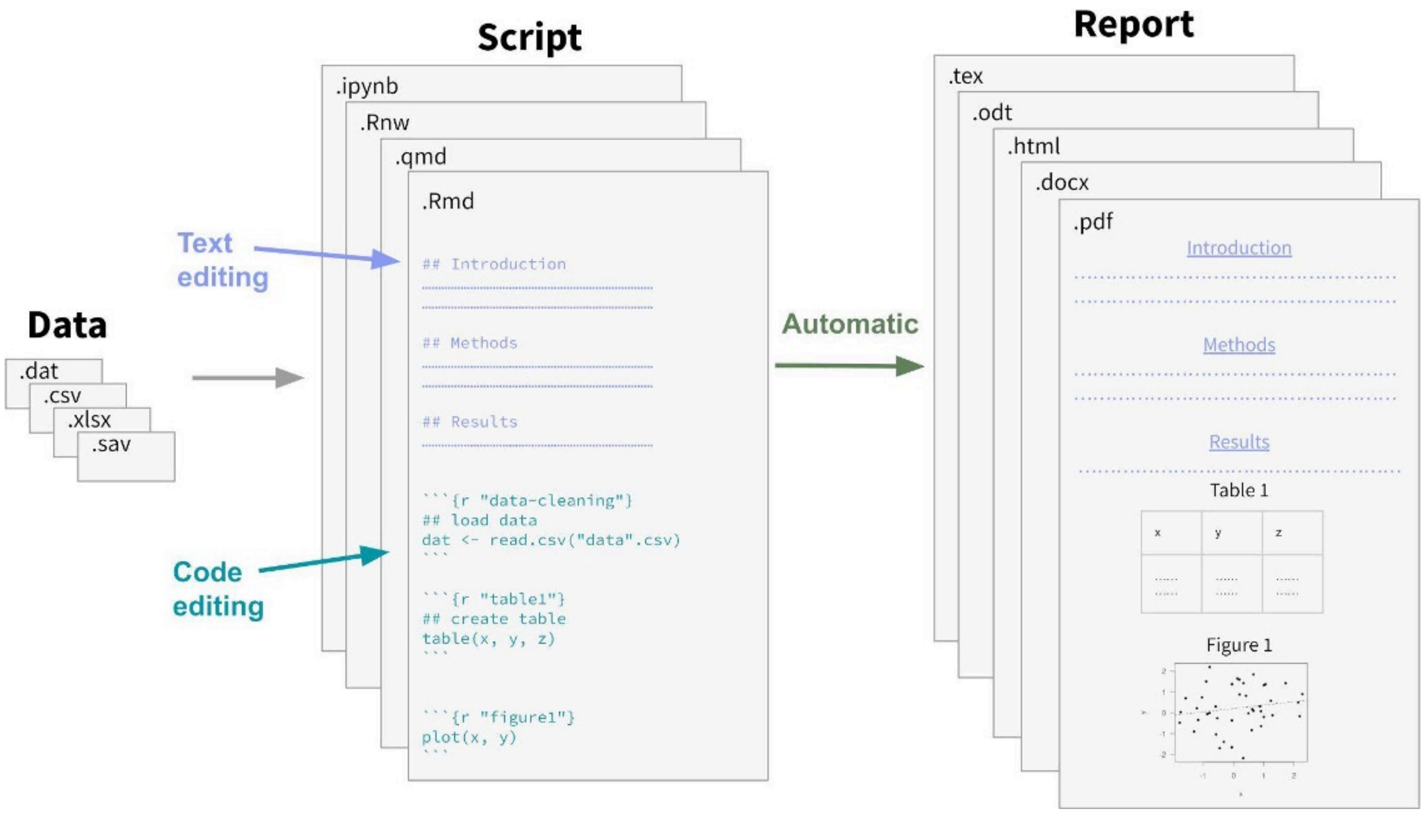
Dynamic reporting

- defines order in one single file
- simplifies documentation with a markup language
- automates generation and processing of code output

A non-dynamic reporting workflow



A dynamic reporting workflow



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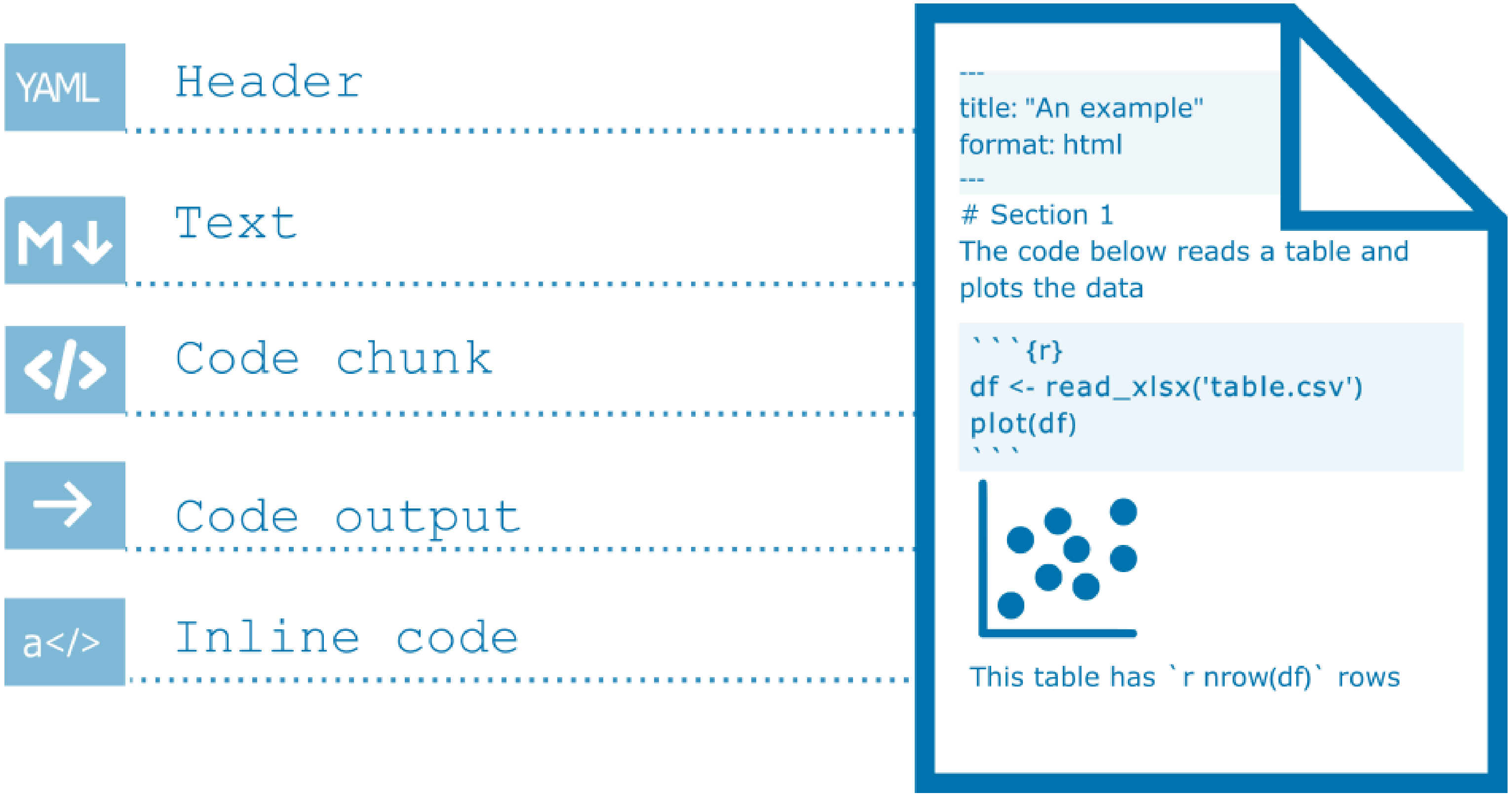
Components of a dynamic report



Common markup languages

HTML

```
<!DOCTYPE html>
<html>
<head>
<title>Document</title>
</head>
<body>
 <h1>Heading Level 1</h1>
 <h2>Heading Level 2</h2>
 <p><strong>Bold text</strong>
and <em>italic text</em> in a
paragraph.</p>

<ol> <li>First item in an ordered
list</li> <li>Second item in an ordered
list</li> </ol>

<ul> <li>Unordered list item</li>
<li>Another unordered list item</li>
</ul> <p><code>Inline code</code>
and</p>

<pre>Block code</pre>

</body>
</html>
```

LaTeX

```
\documentclass{article}
\begin{document}
\section{Heading Level 1}
\subsection{Heading Level 2}

\textbf{Bold text} and \textit{italic
text} in a paragraph.
\begin{enumerate}
  \item First item in an ordered list
  \item Second item in an ordered list
\end{enumerate}

\begin{itemize}
  \item Unordered list item
  \item Another unordered list item
\end{itemize}

\texttt{Inline code} and
\begin{verbatim} Block code \end{verbatim}
\end{document}
```

Markdown

````
# Heading Level 1
## Heading Level 2

**Bold text** and *italic text*
in a paragraph.

1. First item in an ordered list
2. Second item in an ordered list

- Unordered list item
- Another unordered list item

`Inline code` and

```
Block code
```
````

Popular implementations

	Code	Markup	Output formats
knitr .Rnw	R	LaTeX	pdf, tex
R Markdown	R	Markdown	pdf, html, docx, pptx, tex
Jupyter Notebook	Julia, Python, R	Markdown	pdf, html, tex, py, …
Quarto	Julia, Python, R	Markdown	pdf, html, tex, docx, pptx, …

Yihui Xie's blog post: With Quarto coming, is R Markdown going away? No.

The Quarto Render action

Source: https://quarto.org/docs/get-started/hello/rstudio.html

Limitations and Outlook

Limitations

- adds new dependencies
- does not guarantee computational reproducibility
- challenges for collaborative editing of manuscripts

Facilitators

- integrates well with other good practices for computational reproducibility
- growing number of tools for collaborative editing of source documents

Additional Resources

- CRS Primer on dynamic reporting: https://doi.org/10.5281/zenodo.7565735
- Quarto intro tutorial: https://quarto.org/docs/get-started/hello/rstudio.html
- Quarto authoring tutorial: https://quarto.org/docs/get-started/authoring/
- Quarto article layout: https://quarto.org/docs/authoring/article-layout.html
- R4DS chapter on Quarto: https://r4ds.hadley.nz/quarto
- Reproducible manuscripts with Quarto: Slides by Mine Cetinkaya-Rundel
- Quarto/RMarkdown – What's different?: Slides by Ted Laderas

The Center for Reproducible Science

UZH

Center for Reproducible Science

Teaching and training:

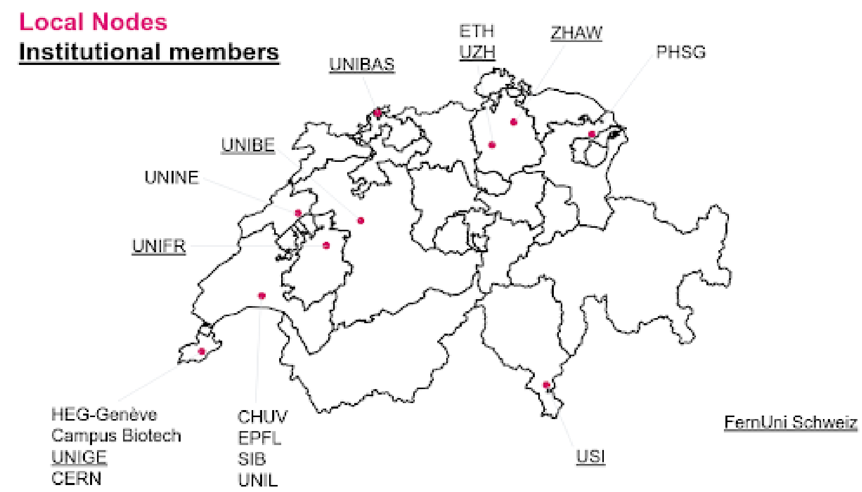
- Good Research Practice courses
- Primers and Reproducibility Notes
- ReproducibiliTea journal club

Research:

- Design and analysis of replication studies
- Meta-research

The Swiss Reproducibility Network

RN CH Swiss Reproducibility Network



Working groups:

- Computational Reproducibility
- Open Research Data
- Preregistration and Registered Reports
- Training
- Research Assessment and Incentives

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